HMC with the even-odd multigrid of RutticeQCD: a walkthrough

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Internal note

This document goes through HMC for Möbius domain-wall fermions, from the path integral to the solves of the force, and shows where the multigrid enters. It is the companion of the MG walkthrough[1], which derives the operator, the even-odd forms, the multigrid and the solve with $D^\dagger$; the MG note leaves HMC to future work. Nothing here is new with respect to the code.

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[1] `docs/mg_background/MG_WALKTHROUGH.tex`; references to it are marked "of the MG walkthrough".

1 What we want to compute

Lattice QCD computes expectation values in the path integral,

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \mathcal{D}U\, \mathcal{D}\bar{\psi}\, \mathcal{D}\psi\; \mathcal{O}\, e^{-S_g[U] - \bar{\psi} D[U] \psi}, \qquad Z = \int \mathcal{D}U\, \mathcal{D}\bar{\psi}\, \mathcal{D}\psi\; e^{-S_g[U] - \bar{\psi} D[U] \psi}, \tag{1.1}$$

with U the gauge links, S_g the gauge action (here the plaquette action at $\beta = 6.0$) and ψ the quark field. The quark fields are Grassmann numbers, and their integral can be done exactly:

$$\int \mathcal{D}\bar{\psi}\, \mathcal{D}\psi\; e^{-\bar{\psi} D \psi} = \det D. \tag{1.2}$$

For two degenerate flavors (up and down) each flavor gives one $\det D$:

$$\begin{aligned} Z &= \int \mathcal{D}U\; e^{-S_g[U]}\, \det D[U]\, \det D[U] \\ &= \int \mathcal{D}U\; e^{-S_g[U]}\, \det(D^\dagger D)[U], \end{aligned} \tag{1.3}$$

where the second line uses that $\det D$ is real, so $(\det D)^2 = \det D^* \det D = \det(D^\dagger D)$. That $\det D$ is real follows from the exact relation $D_- \Gamma_5 D^\dagger = D \Gamma_5 D_-^\dagger$ of Sec. 1.5 of the MG walkthrough, taking determinants ($\det \Gamma_5 = \pm 1$ cancels on both sides):

$$\begin{aligned} \det D_-\, \det D^\dagger &= \det D\, \det D_-^\dagger \\ \det D_-\, (\det D)^* &= \det D\, (\det D_-)^* \\ (\det D)^* &= \det D. \end{aligned} \tag{1.4}$$

The last step divides by $\det D_-$, which is real: since $\Gamma_5 D_- \Gamma_5 = D_-^\dagger$ (Sec. 1.5 of the MG walkthrough),

$$\det D_- = \det(\Gamma_5 D_- \Gamma_5) = \det D_-^\dagger = (\det D_-)^*. \tag{1.5}$$

So the problem is to generate gauge fields U with the probability

$$P[U] \;\propto\; e^{-S_g[U]} \det(D^\dagger D)[U]. \tag{1.6}$$

$\det(D^\dagger D)$ is the hard part: it couples all sites at once, and computing it directly costs $O(V^3)$.

The Pauli–Villars ratio. The domain-wall operator has L_s slices. Only the modes bound to the walls are the physical quark; the modes in the bulk of the fifth dimension are heavy, unphysical, and their determinant diverges as $L_s \to \infty$. They are removed by dividing by the determinant of the same operator at unit mass, the Pauli–Villars operator, whose wall modes are heavy too:

$$\det(D^\dagger D) \;\longrightarrow\; \frac{\det(D^\dagger D)}{\det(D_{\mathrm{PV}}^\dagger D_{\mathrm{PV}})}. \tag{1.7}$$

This is the "Ratio" action.

2 Pseudofermions: a determinant as a Gaussian integral

One complex variable. For a number $a > 0$ and a complex variable $\phi = x + iy$,

$$\begin{aligned} \int \frac{dx\, dy}{\pi}\; e^{-\phi^* a^{-1} \phi} &= \int \frac{dx\, dy}{\pi}\; e^{-(x^2+y^2)/a} \\ &= \frac{1}{\pi} \Big(\int dx\, e^{-x^2/a} \Big) \Big(\int dy\, e^{-y^2/a} \Big) \\ &= \frac{1}{\pi} \sqrt{\pi a}\, \sqrt{\pi a} = a. \end{aligned} \tag{2.1}$$

Many variables. For a Hermitian positive matrix $A = V\Lambda V^\dagger$ (eigenvalues $a_k > 0$), change variables $\phi = V\chi$; the Jacobian of a unitary change is 1:

$$\begin{aligned}\int \mathcal{D}\phi\, e^{-\phi^\dagger A^{-1}\phi} &= \int \mathcal{D}\chi\, e^{-\chi^\dagger V^\dagger V\Lambda^{-1}V^\dagger V\chi} \\ &= \int \mathcal{D}\chi\, e^{-\sum_k |\chi_k|^2/a_k} \\ &= \prod_k a_k = \det A. \end{aligned} \tag{2.2}$$

So a determinant is a Gaussian integral over *bosonic* fields ϕ, the pseudofermions:

$$\det A = \int \mathcal{D}\phi\, e^{-\phi^\dagger A^{-1}\phi}. \tag{2.3}$$

The ratio. Take A such that $\det A$ is our ratio. With $H = \Gamma_5 D$ (whose determinant has the modulus of $\det D$),

$$\begin{aligned} A &= H_{\rm PV}\,(H^\dagger H)^{-1}\,H_{\rm PV}^\dagger, \\ \det A &= \det H_{\rm PV}\,\det(H^\dagger H)^{-1}\,\det H_{\rm PV}^\dagger = \frac{\det(H_{\rm PV}^\dagger H_{\rm PV})}{\det(H^\dagger H)}. \end{aligned} \tag{2.4}$$

Note this is the *inverse* of the ratio; the action uses A^{-1} in the exponent, so

$$\int \mathcal{D}\phi\, e^{-\phi^\dagger A\phi} = \det A^{-1} = \frac{\det(H^\dagger H)}{\det(H_{\rm PV}^\dagger H_{\rm PV})} = \frac{\det(D^\dagger D)}{\det(D_{\rm PV}^\dagger D_{\rm PV})}, \tag{2.5}$$

and the pseudofermion action is

$$S_F[U,\phi] = \phi^\dagger A\phi = \phi^\dagger H_{\rm PV}(H^\dagger H)^{-1}H_{\rm PV}^\dagger\phi. \tag{2.6}$$

The joint distribution of (U,ϕ) is $e^{-S_g - S_F}$; integrating out ϕ gives back $P[U]$.

3 Even-odd: only $\mathcal{K}$ depends on the links

Factor D into block triangular matrices (LDU):

$$\begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix} = \begin{pmatrix} 1 & D_{eo}C_o^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & C_o \end{pmatrix} \begin{pmatrix} 1 & 0 \\ C_o^{-1}D_{oe} & 1 \end{pmatrix}. \tag{3.1}$$

Check by multiplying out the last two factors and then the first:

$$\begin{aligned} \begin{pmatrix} S & 0 \\ 0 & C_o \end{pmatrix} \begin{pmatrix} 1 & 0 \\ C_o^{-1}D_{oe} & 1 \end{pmatrix} &= \begin{pmatrix} S & 0 \\ D_{oe} & C_o \end{pmatrix}, \\ \begin{pmatrix} 1 & D_{eo}C_o^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} S & 0 \\ D_{oe} & C_o \end{pmatrix} &= \begin{pmatrix} S + D_{eo}C_o^{-1}D_{oe} & D_{eo} \\ D_{oe} & C_o \end{pmatrix} = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}. \end{aligned} \tag{3.2}$$

The triangular factors have determinant 1, so

$$\begin{aligned} \det D &= \det S\,\det C_o \\ &= \det\left(C_e\mathcal{K}\right)\det C_o \qquad (\mathcal{K} = C_e^{-1}S) \\ &= \det C_e\,\det C_o\,\det\mathcal{K}. \end{aligned} \tag{3.3}$$

C_e and C_o do not depend on the links (Sec. 1.6 of the MG walkthrough), so they are constants and drop out of $P[U]$. Only $\det \mathcal{K}$ is needed, on the even sites alone, half the lattice. The action becomes

$$S_F = \phi_e^\dagger H_{\rm PV} (H^\dagger H)^{-1} H_{\rm PV}^\dagger \phi_e, \qquad H = \Gamma_5 \mathcal{K}, \quad H_{\rm PV} = \Gamma_5 \mathcal{K}_{\rm PV}, \tag{3.4}$$

and since Γ_5 is Hermitian and unitary, $H^\dagger H = \mathcal{K}^\dagger \Gamma_5 \Gamma_5 \mathcal{K} = \mathcal{K}^\dagger \mathcal{K}$, so with $v_1 = H_{\rm PV}^\dagger \phi_e = \mathcal{K}_{\rm PV}^\dagger \Gamma_5 \phi_e$,

$$\boxed{S_F = v_1^\dagger \, (\mathcal{K}^\dagger \mathcal{K})^{-1} \, v_1} \tag{3.5}$$

This is the action of the code (Bridge++'s `Action_F_alt_Ratio_eo`).

4 The idea of HMC

Sampling U with $P[U] \propto e^{-S[U]}$, $S = S_g + S_F$, by small random changes is hopeless: each accepted change moves one link a little. HMC moves all links at once along a long path, and still samples $P[U]$ exactly.

Fictitious momenta. For every link introduce a momentum $\Pi_\mu(x)$, a traceless Hermitian 3×3 matrix, and the Hamiltonian

$$\mathcal{H}[U, \Pi] = \tfrac{1}{2} \sum_{x,\mu} \operatorname{tr} \Pi_\mu(x)^2 + S[U]. \tag{4.1}$$

The joint distribution $e^{-\mathcal{H}} = e^{-\frac{1}{2}\sum \operatorname{tr}\Pi^2} \, e^{-S[U]}$ is a product: the momenta are Gaussian and independent of U. Sampling (U, Π) from $e^{-\mathcal{H}}$ and forgetting Π samples U from e^{-S}.

One trajectory.

1. Draw Π from its Gaussian and ϕ_e by the heat bath (below).
2. Evolve (U, Π) for a fictitious time τ (here $\tau = 1$) by Hamilton's equations,

   $$\dot{U}_\mu(x) = i \, \Pi_\mu(x) \, U_\mu(x), \qquad \dot{\Pi}_\mu(x) = -F_\mu(x), \tag{4.2}$$

   with the force F the derivative of S with respect to U (below). Exactly solved, these conserve $\mathcal{H}$.
3. The numerical integration does not conserve $\mathcal{H}$ exactly. Accept the new U with probability $\min(1, e^{-\Delta\mathcal{H}})$, $\Delta\mathcal{H} = \mathcal{H}_{\rm end} - \mathcal{H}_{\rm start}$; otherwise keep the old U (the Metropolis test).

Why the result is exact. The test makes the method exact for any step size, provided the integrator has two properties: it is *reversible* (running it with the momenta reversed brings $(U, -\Pi)$ back) and it *preserves volume* in (U, Π) space. Then the proposal $(U, \Pi) \to (U', \Pi')$ is as likely as its reverse, and the acceptance $\min(1, e^{-\Delta\mathcal{H}})$ gives detailed balance with $e^{-\mathcal{H}}$:

$$\begin{aligned} e^{-\mathcal{H}(U,\Pi)} \min\left(1, e^{-[\mathcal{H}(U',\Pi') - \mathcal{H}(U,\Pi)]}\right) &= \min\left(e^{-\mathcal{H}(U,\Pi)}, e^{-\mathcal{H}(U',\Pi')}\right) \\ &= e^{-\mathcal{H}(U',\Pi')} \min\left(1, e^{-[\mathcal{H}(U,\Pi) - \mathcal{H}(U',\Pi')]}\right), \end{aligned} \tag{4.3}$$

which is symmetric in the two states. A small step size only raises the acceptance.

5 The integrator

Leapfrog. Split a step of size ϵ into half a momentum update, a full link update, and half a momentum update:

$$\begin{aligned} \Pi &\leftarrow \Pi - \tfrac{\epsilon}{2} F(U), \\ U &\leftarrow e^{i\epsilon\Pi} U, \\ \Pi &\leftarrow \Pi - \tfrac{\epsilon}{2} F(U). \end{aligned} \tag{5.1}$$

Each of the three updates changes only U or only Π by an amount that depends on the other one, so each is volume preserving (a shear), and the product of shears is too. Reversing Π and running the step again undoes it, update by update. The energy error after a trajectory is $O(\epsilon^2)$.

Omelyan. With a parameter $\lambda \approx 0.193$,

$$\Pi \leftarrow \Pi - \lambda\epsilon F, \quad U \leftarrow e^{i\frac{\epsilon}{2}\Pi} U, \quad \Pi \leftarrow \Pi - (1-2\lambda)\epsilon F, \quad U \leftarrow e^{i\frac{\epsilon}{2}\Pi} U, \quad \Pi \leftarrow \Pi - \lambda\epsilon F. \tag{5.2}$$

It is still reversible and volume preserving, and λ is chosen to make the leading error term about ten times smaller than that of leapfrog, at two force evaluations per step instead of one. $e^{i\epsilon\Pi}$ is evaluated by its Taylor series (order 8 in the runs).

Nested steps. The gauge force is cheap and the fermion force expensive. The runs use two levels: the fermion force with n Omelyan steps, and within each of its link updates the gauge force with 2 steps of its own ("`number_of_steps: [n, 2]`").

How many solves. At the fermion level, consecutive momentum updates of neighboring steps merge into one ($\lambda\epsilon + \lambda\epsilon = 2\lambda\epsilon$). Count the force evaluations of n steps:

$$\underbrace{1}_{\lambda\epsilon \text{ at the start}} + \underbrace{n}_{(1-2\lambda)\epsilon} + \underbrace{n-1}_{2\lambda\epsilon \text{ between steps}} + \underbrace{1}_{\lambda\epsilon \text{ at the end}} = 2n+1. \tag{5.3}$$

Each needs one solve with $\mathcal{K}^\dagger\mathcal{K}$, and the energy at the start and the end one more each:

$$\text{solves per trajectory} = 2n + 3. \tag{5.4}$$

For $n = 20$ this is 43 and for $n = 40$ it is 83, the numbers the runs report.

Step size and stability. The force grows as the quark mass falls (the low modes of $\mathcal{K}$ come closer to zero). If ϵ times the largest force frequency exceeds a threshold, the integrator becomes unstable and $\Delta\mathcal{H}$ explodes rather than growing smoothly. On the dynamical 16^4 configuration at $m = 0.01$, $n = 10$ gave $\Delta\mathcal{H} \approx 10^3$ (every trajectory rejected), $n = 20$ gave $|\Delta\mathcal{H}| \approx 0.05$ (acceptance 0.98), and $n = 40$ gave 0.005.

6 The heat bath for ϕ_e

ϕ_e must be drawn from e^{-S_F} at fixed U. Draw ξ_e Gaussian, $P(\xi_e) \propto e^{-|\xi_e|^2}$, and set

$$\phi_e = H_{\mathrm{PV}} \left(H_{\mathrm{PV}}^\dagger H_{\mathrm{PV}}\right)^{-1} H^\dagger \, \xi_e. \tag{6.1}$$

This is $\phi_e = (H_{\mathrm{PV}}^\dagger)^{-1} H^\dagger \xi_e$, written through the normal equation so that CG can do the solve. Then $S_F = |\xi_e|^2$; the map $\xi_e \mapsto \phi_e$ is linear and does not depend on ϕ_e, so its Jacobian is a constant and

ϕ_e has the distribution e^{-S_F}:

$$\begin{aligned} H_{\rm PV}^\dagger \phi_e &= H_{\rm PV}^\dagger H_{\rm PV} (H_{\rm PV}^\dagger H_{\rm PV})^{-1} H^\dagger \xi_e = H^\dagger \xi_e, \\ S_F &= (H_{\rm PV}^\dagger \phi_e)^\dagger (H^\dagger H)^{-1} (H_{\rm PV}^\dagger \phi_e) \\ &= (H^\dagger \xi_e)^\dagger (H^\dagger H)^{-1} (H^\dagger \xi_e) \\ &= \xi_e^\dagger H\, H^{-1} H^{-\dagger}\, H^\dagger \xi_e \\ &= \xi_e^\dagger \xi_e. \end{aligned} \tag{6.2}$$

This costs one solve with the Pauli–Villars operator ($H_{\rm PV}^\dagger H_{\rm PV}$, well conditioned: CG converges fast).

7 The force

The force is the change of S_F when the links change. Vary U; ϕ_e is fixed. With $\chi = \Gamma_5 \phi_e$, $v_1 = \mathcal{K}_{\rm PV}^\dagger \chi$ and $v_2 = (\mathcal{K}^\dagger \mathcal{K})^{-1} v_1$, and using $\delta(M^{-1}) = -M^{-1}(\delta M) M^{-1}$:

$$\begin{aligned} \delta S_F &= \delta \big[v_1^\dagger (\mathcal{K}^\dagger \mathcal{K})^{-1} v_1 \big] \\ &= (\delta v_1)^\dagger (\mathcal{K}^\dagger \mathcal{K})^{-1} v_1 + v_1^\dagger (\mathcal{K}^\dagger \mathcal{K})^{-1} \delta v_1 + v_1^\dagger\, \delta \big[(\mathcal{K}^\dagger \mathcal{K})^{-1} \big]\, v_1 \\ &= (\delta v_1)^\dagger v_2 + v_2^\dagger\, \delta v_1 - v_1^\dagger (\mathcal{K}^\dagger \mathcal{K})^{-1}\, \delta(\mathcal{K}^\dagger \mathcal{K})\, (\mathcal{K}^\dagger \mathcal{K})^{-1} v_1 \\ &= 2\, \Re\, (\delta v_1)^\dagger v_2 - v_2^\dagger\, \delta(\mathcal{K}^\dagger \mathcal{K})\, v_2 \\ &= 2\, \Re\, (\delta \mathcal{K}_{\rm PV}^\dagger \chi)^\dagger v_2 - v_2^\dagger (\delta \mathcal{K}^\dagger\, \mathcal{K} + \mathcal{K}^\dagger\, \delta \mathcal{K}) v_2 \\ &= 2\, \Re\, \chi^\dagger\, \delta \mathcal{K}_{\rm PV}\, v_2 - 2\, \Re\, (\mathcal{K} v_2)^\dagger\, \delta \mathcal{K}\, v_2. \end{aligned} \tag{7.1}$$

The fourth line uses that $(\mathcal{K}^\dagger \mathcal{K})^{-1}$ is Hermitian, so $v_1^\dagger (\mathcal{K}^\dagger \mathcal{K})^{-1} = v_2^\dagger$, and that $a^\dagger b + b^\dagger a = 2 \Re\, a^\dagger b$. So one force evaluation needs exactly one solve, $v_2 = (\mathcal{K}^\dagger \mathcal{K})^{-1} v_1$; the rest are multiplications. $\delta \mathcal{K}$ comes from the hops in S alone (C_e does not depend on the links): for a variation of one link, only the hops through that link contribute, and the two terms above become a local outer product of v_2, $\mathcal{K} v_2$ and χ at the two ends of the link. With stout smearing the operator sees the smeared links; the force is taken with respect to them and carried back to the thin links by the chain rule through the smearing steps.

8 How precisely to solve

Energy. $\mathcal{H}$ at the start and the end enters $\Delta \mathcal{H}$ directly, and an error there biases the Metropolis test. These solves are done to $|r|^2/|b|^2 < 10^{-24}$.

Force. An error in the force only changes the trajectory, which the test corrects, *as long as the integrator stays reversible*. A solve from a zero initial guess is a fixed function of the links, so reversibility holds to its tolerance; 10^{-16} is ample. (Starting the force solves from the previous solutions, `number_of_initial_guess`, saves iterations but breaks reversibility at the level of the tolerance; it is off by default.)

9 Solving $(\mathcal{K}^\dagger \mathcal{K})^{-1}$

With CG. $\mathcal{K}^\dagger \mathcal{K}$ is Hermitian positive, so CG solves it directly; each iteration costs two applications of $\mathcal{K}$. Mixed precision runs CG in single precision with corrections in double. With Neff's α,

Eq. (2.9) of the MG walkthrough gives $\mathcal{K} = Q_e \mathcal{K}^{(\alpha)} Q_e^{-1}$ and

$$\begin{aligned}
\mathcal{K}^\dagger \mathcal{K} &= Q_e^{-1} \mathcal{K}^{(\alpha)\dagger} Q_e \, Q_e \mathcal{K}^{(\alpha)} Q_e^{-1} \\
&= Q_e^{-1} \, (Q_e \mathcal{K}^{(\alpha)})^\dagger (Q_e \mathcal{K}^{(\alpha)}) \, Q_e^{-1} \\
&= Q_e^{-1} \, (\mathcal{K} Q_e)^\dagger (\mathcal{K} Q_e) \, Q_e^{-1} \qquad (Q_e \mathcal{K}^{(\alpha)} = \mathcal{K} Q_e), \\
(\mathcal{K}^\dagger \mathcal{K})^{-1} &= Q_e \big[(\mathcal{K} Q_e)^\dagger (\mathcal{K} Q_e) \big]^{-1} Q_e,
\end{aligned} \tag{9.1}$$

so CG runs on the normal equation of $\mathcal{K} Q_e$, better conditioned, while the action and the force stay those of $\mathcal{K}$ (on 24^4 at $m = 0.001$: 22% fewer iterations).

With the multigrid: two solves. The multigrid solves with D and $D^\dagger$, not with $\mathcal{K}^\dagger \mathcal{K}$. Invert the product, with $\mathcal{K} = C_e^{-1} S$:

$$\begin{aligned}
(\mathcal{K}^\dagger \mathcal{K})^{-1} &= \mathcal{K}^{-1} \mathcal{K}^{-\dagger} \\
&= (C_e^{-1} S)^{-1} \big[(C_e^{-1} S)^\dagger \big]^{-1} \\
&= S^{-1} C_e \, C_e^\dagger S^{-\dagger}.
\end{aligned} \tag{9.2}$$

So $z = (\mathcal{K}^\dagger \mathcal{K})^{-1} r$ is two solves:

$$\begin{aligned}
w &= C_e^\dagger \, S^{-\dagger} r = C_e^\dagger \, \pi D^{-\dagger} \iota \, r && \text{(with } D^\dagger\text{, MG walkthrough Sec. 8)}, \\
z &= S^{-1} C_e \, w = \pi D^{-1} \iota \, C_e w && \text{(with } D\text{)}.
\end{aligned} \tag{9.3}$$

Each solve stops at 10^{-10}, while the normal equation needs 10^{-16} or 10^{-24}. So the two solves are repeated on the residual of the normal equation (iterative refinement):

$$\begin{aligned}
r^{(k)} &= b - \mathcal{K}^\dagger \mathcal{K} z^{(k)}, \\
z^{(k+1)} &= z^{(k)} + \mathcal{K}^{-1} \mathcal{K}^{-\dagger} r^{(k)},
\end{aligned} \tag{9.4}$$

where $\mathcal{K}^{-1} \mathcal{K}^{-\dagger} r^{(k)}$ stands for the two inexact solves above. If they reduce the residual by a factor ϵ, then $|r^{(k+1)}| \lesssim \epsilon |r^{(k)}|$, and each sweep gains about ten digits; the code allows at most eleven sweeps. With α the solve with D uses $D^{-1} = Q D^{(\alpha)-1}$ (the left form $\tilde{A}_L^{(\alpha)}$); the solve with $D^\dagger$ is unchanged, since $D^{(\alpha)\dagger} = Q D^\dagger$ only rescales its right-hand side.

When the sweeps stall. If the sweeps stop gaining (the coarse space no longer fits the links, as after large link changes), the solve is repeated from the current z with $\mathcal{P}$ and the coarse operators rebuilt for the current links, then also after a bootstrap; if it still stalls, CG finishes it from the current z, with a warning. A trajectory is never lost to the multigrid.

10 What the multigrid keeps along a trajectory

The links change a little at every step of the integrator, and the setup is expensive, so the multigrid is not rebuilt for every solve.

- The test vectors y are kept from solve to solve (on the host between solves) and improved by a bootstrap, from the old y, when the pseudofermions are drawn; the filter runs only for the first trajectory.
- $\mathcal{P}$, the coarse operators and the top of the smoother's interval are kept while the links move and rebuilt for the current links when the GCR iterations per solve have grown by half (`rebuild_iteration_growth` $= 1.5$) or after a bootstrap.
- The cycle is made for each solve from these parts and dropped after it.

A kept coarse space is the coarse space of slightly different links. It is still a good preconditioner; GCR on the operator of the current links keeps the solve exact.

11 Memory (24^4, $m = 0.001$, 20 vectors, levels below)

The memory table and the measurements below were taken with MINRES on the Pauli–Villars factor, before the regularized solve; they have not been repeated with it.

part	GB
HMC itself (links, smeared levels, momenta, pseudofermion, operators)	3.3
multigrid kept between solves ($\mathcal{P}$ 4.75)	6.6
fields of the sweeps	2.6
the solve with $D^\dagger$ on top (GCR, Schur factors, cycle)	6.4
peak (live)	19.1

Mixed CG needs about 9 GB on the same lattice.

12 What the measurements say so far

- On a thermalized dynamical 16^4 configuration (60 trajectories at $m = 0.05$, then 40 at $m = 0.01$ from the quenched one) the multigrid reproduces the trajectories of CG: the plaquette to 10^{-11}, $\Delta\mathcal{H}$ to three digits. This checks that it solves the right equation.
- There CG needs about 300 iterations per solve at $m = 0.01$ and at $m = 0.001$ alike: on a lattice this small the lowest eigenvalues are held up by the finite volume, not by the mass. The multigrid needs about 55 iterations, but one of them (a whole cycle) costs about 24 CG iterations, so it is 3–4 times slower per trajectory. Tuning gains at most 6%.
- On the quenched 24^4 configuration CG needs about 6400 iterations per solve and the multigrid wins, but a trajectory of length 1 there blows up ($\Delta\mathcal{H} \approx 4 \times 10^9$), so it is no test of HMC. There are no data yet that show the multigrid faster in HMC; that needs a volume where CG takes more than about 1300 iterations per solve.

13 Map to the code

step	where
HMC: two solves, sweeps, retries	`src/hmc/dwf_multigrid.rs`; `src/backend/cuda_bridge/hmc_multigrid.rs`
HMC: action, force, CG with α, CG fallback	`src/hmc/even_odd.rs`; `src/hmc/solve.rs`